AI HANDOFF DOCUMENT – NVIDIA LABS iQuHACK 2026

Purpose of this document

This document transfers **strategic intent, technical architecture, and research grounding** for the NVIDIA LABS iQuHACK 2026 challenge to a new AI instance or teammate. It captures *why* we are making certain design choices, not just *what* to implement.

Challenge context

Challenge: NVIDIA LABS – Low Autocorrelation Binary Sequences (LABS)

Core task: Build a **hybrid quantum–classical solver** for LABS where: - A **quantum algorithm generates candidate bitstrings** - Those bitstrings **seed a classical Memetic Tabu Search (MTS)** - The workflow is **GPU-accelerated** (quantum simulation and/or classical search)

The challenge explicitly references a recent academic approach as a recommended template.

The three core papers (DO NOT LOSE THESE)

These papers define the intellectual backbone of the project.

1. Gómez-Cadavid et al. (2025)

"Scaling advantage with quantum-enhanced memetic tabu search for the Low Autocorrelation Binary Sequences problem"

Role in project: - Treated as the **baseline / template** endorsed by the challenge - Defines the core architecture: **quantum seeding → MTS → scaling-style evaluation** - Establishes expected metrics: time-to-solution, success probability, scaling with N

Status: - This method **must be implemented** or approximated - It is **not** the final contribution

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2. Hegade et al. (2022)

"Digitized Counterdiabatic Quantum Optimization" (Phys. Rev. Research)

Role in project: - Provides the **theoretical justification** for why DCQO-style circuits outperform naive QAOA at low depth - Explains how **counterdiabatic (CD) terms suppress diabatic transitions**, improving ground-state tracking

Status: - Used to **justify and motivate** the DCQO-style quantum seeder - Serves as the mechanism-level explanation behind the Cadavid approach

Key takeaway:

> DCQO improves the *distribution* of sampled solutions, not just single-shot outcomes

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3. Sciorilli et al. (2025)

"A competitive NISQ and qubit-efficient solver for the LABS problem" (Pauli Correlation Encoding, PCE)

Role in project: - Introduces **Pauli Correlation Encoding (PCE)** as a qubit-efficient alternative quantum encoding for LABS - Claims strong performance with fewer qubits and shallow circuits

Status: - This is the **primary "twist" paper** used to go *beyond* the challenge template - Implemented as an **alternative quantum seeding engine**

Key design decision (CRITICAL)

We do NOT embed new ideas inside the Cadavid pipeline

Instead: - **Cadavid QE-MTS is treated as one seeding baseline** - New ideas are implemented as **alternative quantum seeders** - All seeders feed the **same classical MTS + GPU-accelerated core**

This preserves: - Clean attribution - Rigorous ablation - Judge credibility

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System architecture (fixed vs variable)

Fixed components (shared across all experiments)

- LABS objective (energy / merit factor)
- Memetic Tabu Search (MTS)
- GPU acceleration strategy
- Runtime budgets
- Evaluation metrics

Variable component: Quantum seeding engine

Each seeder is plug-and-play and interchangeable.

Seeders to compare: 1. **Random seeding** (sanity baseline) 2. **QAOA-based seeding** (canonical baseline) 3. **DCQO-based seeding** (Gómez-Cadavid / Hegade-inspired) 4. **PCE-based seeding** (Sciorilli et al. – main twist)

Architecture:

Quantum seeder → identical MTS (GPU-accelerated) → result

Why this framing is deliberate

Judges and reviewers implicitly ask:

> "If I had to solve LABS tomorrow, which quantum seeding strategy would I choose?"

This structure: - Answers that question directly - Avoids "we tweaked the template" vibes - Matches how hybrid optimization methods are evaluated in the literature

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Evaluation strategy

All seeders are evaluated under **identical conditions**.

Metrics: - Best energy / merit factor vs time - Time-to-threshold (reach a target quality) - Success probability across repeated runs - Diversity of generated seeds (e.g., Hamming distance statistics)

Scaling: - Small set of N values (e.g., N = 20, 30, 40) sufficient for hackathon rigor

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GPU acceleration intent

GPU acceleration is applied to: - Classical MTS (parallel evaluation, population-level operations) - Quantum simulation (CUDA-Q / GPU-backed simulators)

Acceleration is treated as **orthogonal** to the seeding comparison: - Same acceleration path for all seeders - Ensures performance gains are attributed correctly

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Final contribution statement (canonical phrasing)

> The NVIDIA challenge recommends DCQO-based quantum seeding as a template. We treat this as a strong baseline and investigate an additional axis of advantage: **alternative quantum encodings for generating higher-quality and more diverse seeds under identical classical optimization and GPU-accelerated conditions**.

If this document is handed to a new AI or teammate

Immediate priorities: 1. Implement the Cadavid-style DCQO seeder + baseline MTS 2. Modularize seeders behind a clean interface 3. Add PCE as an alternative seeder 4. Run head-to-head comparisons under fixed budgets 5. Produce plots showing time-to-solution and seed quality

If these steps are followed, the project remains aligned with: - The NVIDIA LABS challenge - The cited academic literature - A clear "beyond-the-template" contribution